

Fear of Temperature: Provenance-Aware Research Database and Data-Governance Architecture

Executive architecture decision and standards profile

Research dependency and scope qualification

The Deep Research One and Deep Research Two materials were not actually included in the prompt: the text contains placeholders for the Object Definition, Semantic Taxonomy, Heritage Eligibility Matrix, Inclusion and Exclusion Rules, Ethics Requirements, Historical Periodisation, Anchor Source Registry, Recommended Scope, Minimum Viable Evidence Base, and Direct Evidence versus Proxy Matrix. This report therefore treats the requested entities and evidence gates as binding requirements, but its historical boundaries, geographical priorities, and final affect and heritage vocabularies remain **provisional** until those earlier materials are inserted.

The immediate architectural decision is **MODIFY, then GO**:

Build a thesis-scale implementation using PostgreSQL with JSONB, immutable file storage, explicit provenance tables, controlled review queues, and JSON-LD export. Do not begin with a graph database, a triple store, or a complete implementation of CIDOC CRM.

This decision preserves the distinctions required by the thesis—events, observations, works, expressions, manifestations, archived files, assertions, claims, assessments, surveys, rights and reviews—without imposing the operational burden of maintaining several production database systems. PostgreSQL already supports structured relational constraints, indexed JSONB, full-text search, trigram similarity, row-level security, backup and point-in-time recovery; those capabilities are sufficient for the proposed pilot. ¹

The system should be **relational at its core, graph-shaped in its relations, and RDF-compatible at export**. Reuse, circulation, version, translation and provenance relations should be stored as ordinary relational edge tables. Recursive queries and materialized views can support graph analysis. JSON-LD exports can provide semantic interoperability without requiring the research team to administer a live triple store.

Minimal standards application profile

No single standard covers scientific observations, bibliographic versions, digital preservation, provenance, survey research, public affect and heritage-making. The appropriate strategy is therefore an **application profile**: implement selected concepts from several standards, record the version of each external vocabulary, and avoid claiming full conformance unless every relevant requirement has been tested.

Standard or principle	Thesis-scale use	Implementation level	What should not be attempted
W3C PROV-O	Model acquisition, extraction, transformation, translation, annotation, linkage and publication as activities involving entities and agents	Direct conceptual and export mapping for <code>provenance_event</code> , <code>used</code> , <code>generated</code> , <code>wasAssociatedWith</code> , and derivation relations	A complete OWL reasoning environment
PREMIS	Preservation metadata for archived files: object characteristics, fixity, preservation events, agents and preservation rights	Direct field-level borrowing for <code>archived_item</code> , checksums, formats and preservation events	Full repository-level PREMIS implementation
IFLA LRM	Distinguish <code>artifact_work</code> , <code>artifact_expression</code> , <code>artifact_manifestation</code> and <code>archived_item</code>	Strong conceptual adoption, adapted to non-library media	Treating every web page update as a new abstract work
CIDOC CRM	Crosswalk events, actors, places, information objects, creation, use, custody, claims and assessment activities	Conceptual mapping and optional JSON-LD classes	Rebuilding the database as a complete CRM ontology
Dublin Core Terms	Basic public discovery metadata: title, creator, date, language, type, subject, source, rights and spatial coverage	Public-release profile	Using Dublin Core as the internal research schema
DataCite Metadata Schema	Citation and release metadata for datasets, codebooks and versioned research outputs	Release-stage export; use DataCite 4.7 where applicable	Minting a DOI for every internal record
Schema.org	Search-engine-facing descriptions for public datasets, creative works and media	Lightweight JSON-LD output	Using Schema.org as the authoritative research ontology
SKOS	Affect, temperature, heritage, relation and material-type concept schemes with multilingual labels	Direct controlled-vocabulary implementation	Assuming <code>skos:exactMatch</code> proves historical or semantic identity
GeoNames and national gazetteers	External place identifiers and alternate names	Identifier enrichment, not sole authority	Overwriting historical, Indigenous or contested names with one global label

Standard or principle	Thesis-scale use	Implementation level	What should not be attempted
FAIR	Persistent identifiers, discoverable metadata, machine-readable formats, reusable documentation and workflows	Operational principle for non-sensitive outputs	Interpreting “Accessible” as “open to everyone”
CARE	Collective benefit, authority to control, responsibility and ethics for Indigenous and community-governed data	Binding governance overlay when relevant	Treating CARE as optional descriptive metadata

PROV-O is intended to represent and exchange provenance across systems and can be specialised for domain applications. PREMIS organises preservation metadata around digital objects, events, agents and rights. IFLA LRM is a high-level conceptual model rather than a storage design, making it suitable for borrowing the work-expression-manifestation distinctions without implementing a complete library catalogue. CIDOC CRM is designed for integrating heterogeneous cultural-heritage information, but its breadth makes full implementation disproportionate for a Master thesis. ²

Dublin Core is suitable for concise discovery records and explicitly permits its terms to be used outside RDF contexts, including relational databases and JSON. DataCite’s current Metadata Schema 4.7 was released on 3 March 2026 and is appropriate for the identification and citation of research outputs. Schema.org provides broadly deployed **CreativeWork** and **Dataset** types, while SKOS supplies a stable model for concept schemes, labels and semantic relations. ³

FAIR addresses findability, accessibility, interoperability and reuse, including the workflows and algorithms that produce data. CARE adds a people- and purpose-oriented governance layer centred on collective benefit, authority to control, responsibility and ethics. CARE is not an exception to FAIR; it changes who decides whether and under what conditions FAIR-style reuse is appropriate. ⁴

Executive architecture

The minimum viable architecture has five components:

Component	Recommended implementation	Purpose
Authoritative metadata store	PostgreSQL with JSONB and optional PostGIS	Entities, relationships, assertions, gates, statuses, rights and audit history
Immutable object store	Versioned filesystem, institutional storage or S3-compatible storage	Raw downloads, WARC files, PDFs, images, API responses, audio, video and exports
Reproducible pipeline	Python or R scripts, SQL migrations, Git, lockfiles and container definition	Download, validation, normalisation, extraction, linking and export

Component	Recommended implementation	Purpose
Review interface	Minimal web application or database-backed administrative interface	Evidence display, side-by-side deduplication, gate decisions and adjudication
Publication layer	Curated views plus CSV, JSON, JSON-LD and static media derivatives	Rights-filtered public release

The architecture must distinguish **information identity** from **file identity**. A newspaper article can remain one `artifact_work`, have French and English `artifact_expression` records, appear in several newspaper or website `artifact_manifestation` records, and be represented by multiple `archived_item` files such as WARC, PDF, screenshot and extracted text. This is the central modelling decision that prevents exact duplicates, translations, syndication and later reuse from being destroyed.

Conceptual data model, ERD and data dictionary

Object registry without a universal records table

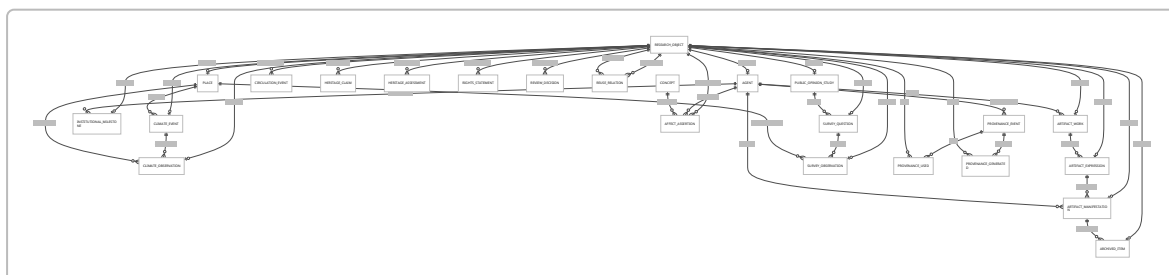
A small `research_object` registry should assign a UUID, type, current status and creation timestamp to every first-class entity. It is **not** a replacement for the twenty-one required tables. It exists only to give provenance, rights, reviews, gate results and generic relations a stable target.

For example:

```
research_object
  object_id
  object_type
  current_status
  created_at
  retired_at
```

`artifact_work`, `climate_event`, `survey_question` and the other domain tables retain their own fields and foreign keys, normally using `object_id` as both primary key and foreign key to `research_object`.

Conceptual ERD



Core entity definitions

Entity	Primary purpose	Key required fields	Principal relationships
<code>climate_event</code>	Bounded heatwave, anomaly period, policy event, report launch or movement event	<code>event_type</code> , name, start/end interval, temporal certainty, scope note	Places, observations, milestones, artefacts
<code>climate_observation</code>	Scientific measurement or statistical climate value	variable, numeric value, unit, observation period, source dataset, dataset version, baseline where relevant, uncertainty	Climate event, place, archived source
<code>institutional_milestone</code>	Formal institutional act or development	type, title, responsible agent, effective/publication date, identifier, documentary source	Agent, place, event, artefact
<code>artifact_work</code>	Abstract intellectual or creative conception	work type, preferred title, creator attribution, creation interval, abstract	Expressions, agents, subjects
<code>artifact_expression</code>	Language, translation, edition, revision, cut or other expression of a work	work ID, language tag, expression type, revision label, translator/editor, source expression	Manifestations, affect assertions
<code>artifact_manifestation</code>	Publication, broadcast, website edition, newspaper issue or distribution format	expression ID, publisher, publication date, edition, channel, canonical identifier	Archived items, circulation
<code>archived_item</code>	Actually preserved bitstream or physical/digital capture	manifestation ID where known, storage URI, MIME type, bytes, checksum, capture time, format validation	Provenance events, rights

Entity	Primary purpose	Key required fields	Principal relationships
<code>agent</code>	Person, organisation, media outlet, government, archive, community or research team	agent type, canonical name, aliases, active interval, stable identifiers	Creation, publication, review, claims
<code>place</code>	Geographic or culturally defined area	preferred name, type, country code where applicable, geometry, uncertainty, external IDs	Events, observations, publication and survey locations
<code>concept</code>	Controlled semantic unit	concept scheme, preferred label, definition, broader/narrower concept, version	Affect, subjects, gates and relations
<code>public_opinion_study</code>	Survey programme, wave or separately sampled study	programme, wave, sponsor, field dates, sample design, mode, weighting documentation	Questions, observations, archived questionnaire
<code>survey_question</code>	Exact questionnaire item in a particular language and version	study ID, exact wording, language, response scale, order, construct, translation provenance	Survey observations and equivalent-question links
<code>survey_observation</code>	Published or computed response estimate	question ID, place, population/group, field period, response category, proportion/estimate, sample size, standard error, weight	Question, study, place
<code>affect_assertion</code>	Evidence-backed interpretation of affect in a source	evidence target, affect class, experiencer/speaker, target, quotation or region locator, method, confidence	Agent, concept, reviewer

Entity	Primary purpose	Key required fields	Principal relationships
<code>circulation_event</code>	Public movement, display, broadcast, sharing, reprint or teaching use	object, circulation type, date, channel, place, audience evidence, source	Manifestation, agent, place
<code>heritage_claim</code>	Claim by an external actor that an object has lasting heritage value	claimant, object, date, claim type, rationale, exact evidence, claimant authority	Agent and source
<code>heritage_assessment</code>	Research-team evaluation under declared criteria	object, rubric version, criterion scores, evidence, result, assessors, confidence	Claims, reviews, gate result
<code>reuse_relation</code>	Later object's evidenced relationship to an earlier object	source, target, relation type, date, evidence, confidence	Works, expressions, manifestations or items
<code>provenance_event</code>	Acquisition or transformation activity	event type, start/end time, software/code version, parameters, agent, outcome	Used and generated objects
<code>rights_statement</code>	Copyright, licence, privacy, ethics and access status	object, copyright status, licence URI, privacy class, display permission, jurisdiction, decision date	Agent, review, public release
<code>review_decision</code>	Human decision with accountable evidence	object, decision type, reviewer, rule version, decision, rationale, timestamp	Gate, status and relation decisions

Supporting relations

The following are necessary even though they are not among the twenty-one primary entities:

Supporting table	Function
<code>source_locator</code>	URL, DOI, catalogue ID, archive ID, API endpoint, call parameters and retrieval time
<code>object_agent_role</code>	Many-to-many creator, publisher, translator, speaker, subject, rights holder and reviewer roles

Supporting table	Function
<code>object_place_role</code>	Event location, observation area, publication location, depicted place and audience location
<code>object_concept</code>	Subject indexing and concept assignments with provenance
<code>external_identifier</code>	DOI, ORCID, ISNI, VIAF, ROR, GeoNames, Wikidata, catalogue and local identifiers
<code>gate_assessment</code>	Gate, result, evidence, rule version, reviewer and timestamp
<code>status_transition</code>	Previous status, new status, authority, rationale and transaction time
<code>concept_scheme</code> and <code>concept_label</code>	Versioned multilingual vocabularies
<code>relation_evidence</code>	Evidence spans, screenshots, quotations or source locators supporting relations
<code>object_version</code>	Supersedes, isVersionOf, effective interval and version label
<code>publication_projection</code>	Public-safe metadata and redacted/derivative references

Controlled vocabulary design

Every controlled value should be a stable identifier rather than free text. Labels can change without changing the identifier. SKOS is appropriate because it separates concept identity from multilingual labels and supports broader, narrower and related relationships. ⁵

The minimum schemes are:

Scheme	Core values
Affect evidence	<code>DIRECT_FEAR</code> , <code>DIRECT_WORRY_ANXIETY</code> , <code>PERCEIVED_THREAT</code> , <code>CONCERN_SERIOUSNESS</code> , <code>RHETORICAL_FEAR_FRAMING</code> , <code>ACCUSATION_OF_PANIC_ALARMISM</code> , <code>INFERRED_AFFECT</code> , <code>ATTENTION_PROXY</code> , <code>BEHAVIOUR_PROXY</code> , <code>NO_AFFECT_EVIDENCE</code>
Heritage status	<code>CULTURAL_RECORD</code> , <code>HERITAGE_CANDIDATE</code> , <code>COMMUNITY_RECOGNISED</code> , <code>INSTITUTIONALLY_RECOGNISED</code> , <code>DISPUTED</code> , <code>NOT_ASSESSED</code> , <code>INSUFFICIENT_EVIDENCE</code>
Reuse relation	<code>SAME_AS</code> , <code>VERSION_OF</code> , <code>TRANSLATION_OF</code> , <code>SYNDICATED_FROM</code> , <code>REPUBLISHES</code> , <code>QUOTES</code> , <code>EXCERPTS</code> , <code>DERIVES_FROM</code> , <code>REMISES</code> , <code>DEPICTS_SAME_EVENT</code> , <code>RESPONDS_TO</code> , <code>COMMEMORATES</code> , <code>REUSED_IN</code>
Date certainty	<code>EXACT</code> , <code>DAY_UNCERTAIN</code> , <code>MONTH_ONLY</code> , <code>YEAR_ONLY</code> , <code>APPROXIMATE</code> , <code>INTERVAL</code> , <code>BEFORE</code> , <code>AFTER</code> , <code>UNKNOWN</code>

Scheme	Core values
Translation method	ORIGINAL, HUMAN_PROFESSIONAL, HUMAN_RESEARCHER, COMMUNITY_VALIDATED, MACHINE, MACHINE_POST_EDITED, UNKNOWN
Review result	PASS, FAIL, REVIEW, NOT_APPLICABLE, SUPERSEDED
Release level	METADATA_ONLY, THUMBNAIL, EXCERPT, FULL_TEXT, FULL_MEDIA, RESTRICTED, EMBARGOED, NO_PUBLIC_RELEASE
Confidence	Numeric probability where calibrated, plus ordinal LOW, MEDIUM, HIGH; never substitute ordinal confidence for evidence
Agent role	Creator, publisher, speaker, translator, editor, collector, claimant, rights holder, custodian, reviewer, community authority
Place role	Event location, measurement location, publication location, depicted location, collection location, survey population

A concept assignment must carry `scheme_version`, `assigned_by`, `assignment_method`, `assigned_at`, and preferably a `review_decision_id`. This makes a classification reproducible even after the vocabulary changes.

Evidence gates, status machine and substantive research protocols

Evidence-gate decision tree

The gates are sequential for promotion, but failure does not normally erase a record. A failed record should remain as a source lead, excluded record or quarantined item with its audit history intact.

Gate	Required fields	Automatic rules	Human rules	Failure treatment	May proceed?
G0 Technical Integrity	Storage URI, retrieval time, checksum, byte size, MIME type, original copy indicator	File exists; checksum recomputes; bytes greater than zero; encoding or format test completes	Inspect malformed, encrypted, truncated or unsupported files	Retry acquisition; quarantine corrupt copy; preserve failed copy if evidentially relevant	Only <code>PASS</code> ; if fails, record remains in raw
G1 Provenance	Creator/publisher where available, date or uncertainty, source URL/identifier, source-chain note	Stable ID resolves; URL and retrieval timestamp present; archival capture linked	Judge source authority, screenshot chain, attribution conflicts and authenticity	<code>SOURCE_LEAD</code> , <code>NEEDS_REVIEW</code> or <code>QUARANTINED</code> ; do not promote as verified	Only <code>PASS</code> to record

Gate	Required fields	Automatic rules	Human rules	Failure treatment	May proceed?
G2 Scope Relevance	Scope concept, date/interval, geography, material type, relevance evidence	Controlled terms and date/ geography bounds; temperature term may trigger review but not pass	Confirm substantive relation to temperature, warming, heat, anomaly or threshold	Retain reference-only contextual source or mark excluded with reason	PASS ; selected records may enter REFERENCE_0
G3 Affect Evidence	Affect class, exact quotation/ question/visual region, speaker or respondent, target, method, confidence	Detect missing evidence; validate allowed class; keyword rules can only nominate candidates	Determine direct expression versus proxy; inspect negation, quotation, sarcasm and translated meaning	Classify as NO_AFFECT_EVIDENCE or review; never fabricate direct fear	Records may proceed supported class analysis
G4 Sociality	Circulation event, audience/ community evidence, public channel, interaction or collective-action evidence	Count known manifestations and circulation links; detect absence of evidence	Distinguish mere publication from demonstrated reception, use or participation	Remain cultural record; no claim of social impact	Heritage promotion requires PASS review
G5 Heritage-Making	Preservation, reuse, commemoration, exhibition, educational use, community/ institutional claim or intergenerational evidence	Count linked claims, archive accessions and reuse events; no automatic heritage recognition	Evaluate durability, value attribution, identity and memory; separate research assessment from external recognition	Remain cultural record; heritage status INSUFFICIENT_EVIDENCE	Only PASS can heritage candidate

Gate	Required fields	Automatic rules	Human rules	Failure treatment	May proceed?
G6 Rights and Ethics	Copyright status, licence, privacy level, personal-data assessment, community restrictions, display permission, takedown route	Missing-rights flag; licence URI validation; minor/sensitive flags; restricted-source rules	Contextual privacy, vulnerable community, Indigenous authority, consent and harm assessment	Internal-only, redact, embargo, metadata-only or quarantine	Internal use m under approva blocked unless
G7 Research Quality	Required-field score, source availability, confidence, conflicts, reviewer requirement, reproducibility record	Completeness and provenance coverage; unresolved-conflict count; transformation reproducibility	Resolve contested classifications; determine double-review need	NEEDS_REVIEW , return to curator, or retain as research-only	Only PASS to release
G8 Public Release	Release level, rights basis, privacy treatment, community permission, quality approval, publication version	Public projection contains no restricted fields; release artefact checksum; approval not expired	Final review of excerpts, thumbnails, de-identification and context	No release, metadata-only, embargo or revised public derivative	Only explicit P PUBLIC_RELE

A gate's **PASS** means “the decision is adequately evidenced,” not “the phenomenon is present.” For example, G3 may pass with **NO_AFFECT_EVIDENCE** because the researcher has reliably established the absence of affect evidence. G5 may pass as an assessment process while concluding **INSUFFICIENT_EVIDENCE**; that conclusion must not trigger heritage status.

Every gate record should contain:

```

gate_assessment_id
object_id
gate_code
result
rule_set_id
rule_version
automatic_checks_json
evidence_object_ids
reviewer_agent_id
reviewed_at

```

decision_reason
supersedes_assessment_id

Automated checks must be rerunnable. Human decisions must be immutable once issued; corrections create a superseding decision rather than overwriting the earlier one.

Affect classification protocol

The thesis must distinguish linguistic or visual evidence of emotion from indicators that merely correlate with attention or concern.

Class	Minimum evidence
<code>DIRECT_FEAR</code>	First-person or clearly attributed use of fear, terror, dread, fright or a validated equivalent whose target includes temperature, warming or extreme heat
<code>DIRECT_WORRY_ANXIETY</code>	Explicit worry, anxiety, unease or validated equivalent
<code>PERCEIVED_THREAT</code>	Explicit claim that the phenomenon threatens life, safety, identity, livelihood or future
<code>CONCERN_SERIOUSNESS</code>	Survey or statement about concern, seriousness or priority without direct fear language
<code>RHETORICAL_FEAR_FRAMING</code>	Speaker or author invokes fear as a persuasive frame, including encouragement, warning or dramatisation
<code>ACCUSATION_OF_PANIC_ALARMISM</code>	Claims that others are panicking, fearmongering, alarmist or hysterical
<code>INFERRED_AFFECT</code>	Researcher inference from context, gesture or imagery; requires a written interpretive rationale and lower default confidence
<code>ATTENTION_PROXY</code>	Search volume, media volume, views, shares or salience without emotional evidence
<code>BEHAVIOUR_PROXY</code>	Avoidance, purchasing, relocation, protest or protective behaviour without direct emotional testimony
<code>NO_AFFECT_EVIDENCE</code>	Relevant to climate or temperature, but no supportable affect classification

The following records **must never be labelled fear from keywords alone**:

- headlines containing “fear,” “panic,” “alarm” or “terror” without reading the body;
- documents quoting, denying, mocking or criticising someone else’s fear;
- search trends and engagement metrics;
- attendance at protests;
- heat warnings, emergency declarations or adaptation policies;
- behavioural responses such as air-conditioner purchases or relocation;
- images with red colour palettes, flames or distressed faces;
- machine translations in which the relevant affect term has not been checked against the original;

- surveys measuring seriousness, concern or threat but not fear;
- reports describing mortality or risk without recording anyone's emotional response.

The `affect_assertion` should record the evidence at the smallest meaningful unit: character offsets for text, timestamp intervals for audio/video, region coordinates for images, or question-and-response identifiers for surveys. A title can be evidence only when the title itself is the public expression being studied, not as a shortcut for classifying the underlying document.

Heritage assessment protocol

Heritage is not an inherent database field. It is a status produced through claims, selection, preservation, reuse, memory practices and institutional or community authority.

A research-team heritage assessment should score six dimensions on a declared ordinal scale, for example zero to two:

Dimension	Zero	One	Two
Deliberate preservation	No evidence	Incidental or platform preservation	Archive, collection or community stewardship
Temporal persistence	One-time appearance	Later rediscovery or isolated reuse	Sustained or repeated reuse
Value attribution	None	Researcher-inferred significance	Explicit community or institutional claim
Collective memory	None	Retrospective historical framing	Repeated commemoration, testimony or identity linkage
Transmission	None	Reuse by same generation/context	Educational or intergenerational transmission
Transformative reuse	None	Quotation/ republication	Adaptation, memorialisation, reinterpretation or ritualised reuse

Recommended decision rules are:

CULTURAL_RECORD:

G0–G3 adequately resolved.

HERITAGE_CANDIDATE:

G5 evidence exists;
at least one direct preservation/value-attribution indicator;
total rubric score at or above a pre-registered threshold;
human review completed.

COMMUNITY_RECOGNISED_HERITAGE:

explicit claim by a relevant community authority or sustained community practice;

claim evidence stored;
community terminology and restrictions respected.

INSTITUTIONALLY_RECOGNISED_HERITAGE:

formal accession, listing, exhibition, curriculum inclusion, archival designation,
commissioned commemoration or equivalent institutional act.

DISPUTED:

credible actors contest significance, ownership, interpretation or authority;
competing claims remain separately represented.

The following are **not sufficient heritage evidence on their own**:

- high share or view counts;
- broad media coverage;
- virality;
- a single archive capture generated automatically;
- inclusion in the research database;
- researcher interest;
- short-term reuse during the same news cycle;
- multiple syndicated manifestations of one newswire text;
- formal publication by a respected institution.

Social circulation can be evidence for G4 and can support a heritage assessment, but it cannot substitute for preservation, persistence, value attribution, memory or transmission.

A `heritage_claim` must remain distinct from `heritage_assessment`. A museum accession, community statement or archive catalogue description is a claim or recognition by an external actor. A thesis coding decision is a research assessment. The latter must never be presented as official designation.

Record-status machine

Status	Definition and required evidence	Typical incoming transitions	Permitted outgoing transitions	Ap pu
<code>REFERENCE_ONLY</code>	Contextual source not intended as a full analytic record	Ingestion or failed scope promotion	<code>SOURCE_LEAD</code> , <code>CANDIDATE_RECORD</code> , <code>EXCLUDED</code>	Cu me no int

Status	Definition and required evidence	Typical incoming transitions	Permitted outgoing transitions	Ap pu
SOURCE_LEAD	URL, citation or mention requiring acquisition and verification	Discovery	CANDIDATE_RECORD , REFERENCE_ONLY , EXCLUDED , QUARANTINED	Cu pu ev dis
CANDIDATE_RECORD	Raw item acquired; G0 passed or under final review	Source lead	PROVENANCE_VERIFIED , NEEDS_REVIEW , QUARANTINED , EXCLUDED	Cu no int
PROVENANCE_VERIFIED	G0 and G1 passed	Candidate	VERIFIED_CULTURAL_RECORD , NEEDS_REVIEW , RETRACTED	Re lim me be aft
VERIFIED_CULTURAL_RECORD	Scope verified and source chain adequate; affect may be absent	Provenance verified	HERITAGE_CANDIDATE , RESEARCH_ONLY , PUBLIC_RELEASE_APPROVED , NEEDS_REVIEW	Do rev
HERITAGE_CANDIDATE	Research rubric supports candidacy; no claim of formal recognition	Verified cultural record	Community or institutional recognition, NEEDS_REVIEW , RESEARCH_ONLY	Tw re pu m ca
COMMUNITY_RECOGNISED_HERITAGE	Relevant community has explicitly conferred or practised durable value	Heritage candidate	PUBLIC_RELEASE_APPROVED , RESEARCH_ONLY , NEEDS_REVIEW , RETRACTED	Co au ev re

Status	Definition and required evidence	Typical incoming transitions	Permitted outgoing transitions	Ap pu
INSTITUTIONALLY_RECOGNISED_HERITAGE	Formal institutional recognition or accession is documented	Heritage candidate	PUBLIC_RELEASE_APPROVED , RESEARCH_ONLY , NEEDS_REVIEW , RETRACTED	Re ve ins so
RESEARCH_ONLY	Analytically usable but restricted, incomplete for public release, or ethically sensitive	Any verified state	PUBLIC_RELEASE_APPROVED , NEEDS_REVIEW , RETRACTED	Re an rig
PUBLIC_RELEASE_APPROVED	G8 passed for a named release version and level	Verified or recognised state	NEEDS_REVIEW , RETRACTED	Rig ap rel ma
NEEDS_REVIEW	Conflict, missing evidence, low confidence or expired decision	Any active state	Return to previous or revised valid state; QUARANTINED	As rev pr wh un
QUARANTINED	Technical, authenticity, malware, privacy or serious ethical problem	Candidate or active state	NEEDS_REVIEW , RESEARCH_ONLY , EXCLUDED , RETRACTED	Se rev pu
EXCLUDED	Outside scope or unusable under declared rules, with reason retained	Lead, candidate, review	NEEDS_REVIEW only through formal reopening	Cu ex me be

Status	Definition and required evidence	Typical incoming transitions	Permitted outgoing transitions	Ap pu
RETRACTED	Previously accepted decision or release withdrawn	Any promoted state	Replacement record or corrected state through superseding decision	Re pu to re

All transitions are reversible through a new decision, not database deletion. Every transition must retain the previous state, approving agent, evidence, rule version, timestamp and reason. A record's current status is a convenience field; the authoritative history is `status_transition`.

Survey harmonisation protocol

Cross-national survey comparison requires more than matching variable names. Cross-cultural survey guidance treats questionnaire design, translation, harmonisation, sampling, processing, adjustment and dissemination as parts of a connected survey lifecycle. AAPOR likewise emphasises methodological disclosure, including exact wording, population, sampling, mode, weighting and fieldwork. ⁶

The unit of harmonisation must be the **question version**, not an abstract construct alone. For every item, preserve:

```
exact wording
introductory text
question order
routing and filter conditions
response labels and numeric codes
showcard or visual scale
original language
source questionnaire
translated version
translation documentation
fieldwork dates
mode
sample design
target population
weight variables
missing-value definitions
derived-variable code
```

Questions should be linked using a relation vocabulary rather than forcibly collapsed:

Relation	Meaning
IDENTICAL_WORDING	Same text and response options in the same language
OFFICIAL_TRANSLATION_OF	Approved language version within the same study

Relation	Meaning
<code>CONSTRUCT_EQUIVALENT</code>	Intended to measure the same construct, with documented differences
<code>PARTIALLY_COMPARABLE</code>	Some comparisons defensible after restriction or recoding
<code>NOT_COMPARABLE</code>	Different target, reference period, scale or framing
<code>SUPERSEDES_QUESTION</code>	Revised item in a later wave
<code>DERIVED_FROM_QUESTION</code>	Recoded or calculated variable

A survey item asking whether climate change is “a serious problem” should normally map to `CONCERN_SERIOUSNESS`, not `DIRECT_FEAR`. An item asking “How worried are you about climate change?” supports direct worry, but not necessarily fear. ESS Round Eight, for example, included a dedicated module on climate beliefs and concern, while Eurobarometer provides repeated cross-national climate surveys; both are useful pilot sources precisely because the questionnaire and methodological context can be retained. ⁷

Published percentages must not be merged across waves unless population, mode, field dates, weighting, wording and response categories are sufficiently comparable. Store the original estimate and response category first; create harmonised estimates only in the analytical layer, with code and assumptions.

Multilingual strategy

Use BCP 47 language tags for language, script and regional variants where known. BCP 47 is designed to identify languages in information objects and supports combinations such as language, script and region. ⁸

For every expression or evidence span, record:

```
language_declared
language_detected
language_detection_model
detection_confidence
script
original_text
normalised_text
transliteration
translation_text
translation_method
translator_agent_id
translation_model_and_version
review_status
untranslated_affect_term
semantic_equivalence_note
```

The original text is authoritative. A translation is a new `artifact_expression` or a research-generated derivative, not a replacement string. Machine translations must retain model name, version

or snapshot, parameters where available, date and source expression. Human corrections create another version.

Multilingual embeddings such as LaBSE can generate cross-language duplicate or translation candidates; LaBSE was developed for language-agnostic sentence representation and supports many languages. Such models are appropriate for candidate generation, not final identity decisions. Their performance is uneven across language families, dialects and low-resource languages, and later work continues to document language-dependent behaviour in multilingual representations. ⁹

Affect lexicons should therefore contain:

- language-specific preferred and alternative labels;
- morphology and common inflections;
- literal glosses;
- culturally situated definitions;
- false friends and negation patterns;
- rhetorical and idiomatic uses;
- source citations;
- community or language-expert validation state.

Cross-language coding should be validated within each major language stratum. A high aggregate accuracy can conceal failure in a smaller corpus.

Raw-to-public pipeline, cleaning and provenance model

Data layers

Layer	Contents	Mutation policy	Promotion requirement
<code>raw_source</code>	Original files, API responses, WARC captures, PDFs, images, subtitles, manifests and acquisition logs	Append-only; never overwritten	G0 technical checks
<code>normalized</code>	Parsed text, standard dates, units, language tags, canonical names and structured source metadata	New version for changed rules	Provenance event linking raw input to output
<code>linked_entities</code>	Agent, place, work/ expression/manifestation, event and version links; candidate and confirmed relations	Relation assertions are versioned	Linkage decision and confidence
<code>curated_research</code>	Records that satisfy scope and quality gates for specified analyses	Curated snapshots by release	G1–G7 as applicable

Layer	Contents	Mutation policy	Promotion requirement
<code>analytical</code>	Time series, term counts, affect models, networks, survey harmonisation and heritage indicators	Rebuildable; never treated as primary evidence	Code version, inputs, parameters and validation
<code>public_release</code>	Rights-cleared metadata, excerpts, thumbnails, public media and documentation	Immutable numbered releases	G8 approval

The raw layer should preserve both the acquired bytes and acquisition context. A downloaded web page may require HTML, HTTP headers, request URL, redirect chain, capture timestamp, screenshot, extracted text and possibly a WARC. A changed web page is a new archived item, even when the canonical URL remains the same.

Mandatory provenance chain

Every derived object must be traceable to:

```
raw archived_item
source URL or stable identifier
retrieval timestamp
checksum and checksum algorithm
acquisition tool and version
transformation activity
transformation code commit or release
parameters and environment
rule-set version
generated object
review decision where applicable
publication status and release version
```

A representative chain is:

```
Source URL
-> acquisition activity
-> raw HTML archived_item, SHA-256 A
-> text extraction activity, code commit C1
-> normalised text archived_item, SHA-256 B
-> language-identification activity, model L1
-> expression metadata version E2
-> affect-coding activity, rules AF-1.3
-> affect_assertion AA7
-> reviewer decision RD18
-> curated snapshot CS4
```

```
-> redaction activity, rules PUB-1.1
-> public JSON record PJ4
```

PROV-O provides the conceptual pattern of entity–activity–agent relationships, while PREMIS provides preservation-oriented event, fixity, object and rights semantics. ¹⁰

Version model

Four version dimensions must remain separate:

Dimension	Example
Source version	Newspaper article updated at 14:00 after original 09:00 publication
File version	New WARC capture or replacement scan
Metadata version	Corrected creator, date or language
Research interpretation version	Affect code changed after adjudication

Recommended fields are:

```
version_number
version_label
valid_from
valid_to
recorded_at
supersedes_id
superseded_by_id
change_reason
changed_by
source_version_identifier
schema_version
rule_version
```

Use **transaction time** and **valid time** where meaningful. `recorded_at` describes when the database learned something; `valid_from` and `valid_to` describe when the assertion is considered true in the source domain.

Cleaning rules

Time

Use ISO 8601 for exact timestamps and EDTF-compatible strings for uncertain or approximate historical dates. EDTF was created to represent features such as uncertain, approximate and interval dates that are difficult to express with a single exact timestamp. ¹¹

Input condition	Storage rule
Exact publication date	<code>publication_start = publication_end</code> ; certainty <code>EXACT</code>
Month known	Store first and last possible day plus lexical value such as <code>2003-08</code>
Approximate year	Interval plus original string and certainty <code>APPROXIMATE</code>
"Before 1950"	Open interval endpoint and original statement
Event spans several days	Start and end dates; never collapse to publication date
Fieldwork spans weeks	Separate <code>fieldwork_start</code> and <code>fieldwork_end</code>
Retrieval	UTC timestamp plus local offset where available
Historical calendar	Original calendar value, calendar system and normalised proleptic-Gregorian interpretation
Conflicting dates	Separate assertions with sources; no silent selection

Never use one generic `date` field for event date, publication date, broadcast date, fieldwork date, accession date and retrieval date.

Geography

A place record should preserve preferred, alternate, historical, endonymic, exonymic and Indigenous names. GeoNames can supply global identifiers and alternate names, but it should be supplemented by national gazetteers, archival authorities and community-governed terminology. GeoNames assigns stable web identifiers to millions of toponyms, making it useful for linking but not sufficient for culturally or historically contested geography. ¹²

Store:

```

place_id
preferred_name
name_authority
place_type
country_code
admin_hierarchy
geometry
coordinate_reference_system
spatial_uncertainty_m
valid_time
historical_name
external_identifiers
sensitivity_class

```

Event location, observation location, publication location, creator location, depicted location and collection location are different relationship roles. A transnational event should link to multiple places or a geometry rather than being assigned arbitrarily to the publisher's country.

Sensitive locations, including culturally restricted sites or locations associated with vulnerable people, should have an internal geometry and a public generalised geometry.

Temperature and climate observations

Use a controlled variable vocabulary and an unambiguous unit code. UCUM is designed for machine communication of scientific units and can represent Celsius and Fahrenheit without relying on ambiguous text labels. ¹³

Required distinctions are:

Dimension	Required values or fields
Measurement type	Observed, reconstructed, reanalysis, forecast, projection
Quantity	Absolute temperature, anomaly, heat index, wet-bulb temperature, humidity, duration
Statistic	Instantaneous, daily maximum, daily minimum, mean, percentile, count above threshold
Spatial basis	Station, grid cell, city, region, national average, global mean
Temporal basis	Timestamp, day, month, season, year, event interval
Anomaly baseline	Start year, end year, dataset's baseline label
Dataset	Name, provider, edition/version, download date and variable identifier
Uncertainty	Standard error, confidence interval, ensemble range or provider's stated uncertainty
Conversion	Original value/unit plus converted value/unit and conversion-code version

A 40°C local maximum cannot be compared directly to a 1.5°C global anomaly. An anomaly without its baseline is incomplete. Forecast and observed values must never occupy the same variable without a measurement-type discriminator.

Language and text

Preserve original Unicode bytes where possible, original encoding declaration, decoded text, normalised text and transformation history. Unicode normalisation, whitespace cleanup, boilerplate removal and OCR correction should each be separate or at least separately logged transformations.

Do not remove punctuation, diacritics, emoji, hashtags or casing from the authoritative text. Search-normalised representations may remove or transform them, but those are derivatives. Transliteration is also a derivative and must not replace the source script.

Survey data

Store original categories and codes before harmonised categories. Record:

```
question order
filters
response scale
coding direction
missing codes
unweighted n
weighted estimate
weight variable
design variables
mode
sample frame
target population
field dates
country and subnational coverage
construct code
comparability decision
```

A percentage without a denominator, sample definition and question version should fail the survey quality check.

Agents and organisations

Store a canonical local label but do not force uncertain entities to match an external authority. Recommended identifiers include ORCID for researchers, ROR for organisations, ISNI or VIAF for public creators where applicable, and local archive identifiers.

Organisation hierarchy requires time-bounded relationships because departments, ministries, publishers and media groups change names and ownership. Store aliases as assertions with source and valid interval. An uncertain person identity should receive a provisional agent record and `POSSIBLY_SAME_AS` relation, not an automatic merge.

Preservation package

For every release, create a preservation package containing:

```
database schema migration
controlled-vocabulary exports
data dictionary
gate and status rule sets
checksums manifest
code repository snapshot
dependency lockfile
container definition
raw-source inventory
curated data export
```

rights and restriction inventory
README and reproducibility instructions

Use SHA-256 or a stronger supported hash for fixity rather than SHA-1; NIST recommends SHA-256 at minimum for interoperable hash applications. ¹⁴

Preservation derivatives should favour well-documented, widely supported formats while retaining original files. The Library of Congress Recommended Formats Statement evaluates formats by sustainability and long-term accessibility and identifies formats such as PDF/A, structured XML and well-documented dataset formats as appropriate in relevant content categories. ¹⁵

Deduplication, entity resolution and validation

Multi-stage linkage protocol

Duplicate detection must distinguish bit-level identity, textual identity, bibliographic identity, intellectual-work identity and historical relationship.

Stage	Process	Output
Deterministic matching	SHA-256, DOI, catalogue ID, archive identifier, canonical URL, exact normalised-text hash	High-confidence candidate or exact binary/text identity
Candidate generation	Blocking by date/publisher/language; title similarity; n-grams; MinHash; perceptual image hash; named entities; shared quotations; multilingual embeddings	Manageable candidate pairs
Probabilistic scoring	Weighted features with calibrated match likelihood and relation-specific models	Match, non-match and clerical-review zones
Human adjudication	Side-by-side files, source metadata, text diffs, image comparison, timeline and provenance	Confirmed relation and confidence
Link without destruction	Store typed relation and retain every appearance	Network and circulation evidence preserved

The Fellegi–Sunter framework provides a principled basis for combining several comparison features into match, non-match and possible-match decisions. MinHash can efficiently estimate document resemblance for candidate generation, and PostgreSQL's trigram facilities support indexed fuzzy text comparison. None of these establishes historical identity by itself. ¹⁶

Required relationship distinctions

Phenomenon	Correct treatment
Exact binary duplicate	Two acquisition/custody records may point to one bitstream fingerprint; do not erase retrieval context
Exact text duplicate	Link <code>SAME_TEXT_AS</code> ; manifestations may remain different

Phenomenon	Correct treatment
URL duplicate	Canonicalise for comparison, but retain observed URL, redirects and retrieval events
Mirrored copy	<code>MIRRORS</code> or <code>REPUBLISHES</code> ; preserve host and capture
Syndicated news	Shared work or source expression plus distinct manifestations and <code>SYNDICATED_FROM</code>
Same text, different headline	Usually same expression with manifestation-specific title; review editorial changes
Revised article	New expression or version, linked <code>VERSION_OF</code>
Translation	New expression, linked <code>TRANSLATION_OF</code>
Republication	New manifestation or expression depending on editorial changes
Excerpt or quotation	Separate work/expression with <code>EXCERPTS</code> or <code>QUOTES</code>
Derivative work	Separate work linked <code>DERIVES_FROM</code>
Remix or meme	Separate work and manifestations; preserve reuse chain
Same speech in several outlets	One speech work/expression, multiple manifestations; publication edits may create expressions
Survey question across waves	Distinct question versions linked by comparability relation
Same event, different coverage	Different artefact works linked <code>DEPICTS_SAME_EVENT</code>
Visually similar historical objects	Separate works unless creation identity is proven

Objects that must never be automatically merged

The following require human review even when similarity is very high:

- distinct historical photographs or artworks;
- people with the same or similar names;
- organisations across mergers, renamings or parent-subsidiary changes;
- places with contested, historical or reused names;
- translations;
- revised or updated articles;
- syndicated articles with local editorial additions;
- survey questions across languages or waves;
- speeches transcribed or edited by different publishers;
- memes, remixes and derivative visual works;
- community heritage claims and institutional heritage claims;
- climate observations from different dataset versions;
- archived items acquired at different times;
- records with different rights or community-access conditions.

An exact checksum proves only that the bytes are identical. It does not prove that two acquisition events, custody contexts, URLs or legal conditions are identical. Conversely, different bytes do not imply different intellectual works.

Probabilistic features and thresholds

A relation-specific model should score features such as:

```
title trigram similarity
title token overlap
author or speaker agreement
publisher agreement
publication-date distance
language agreement
named-entity overlap
shared quotation length
normalised-text Jaccard similarity
MinHash similarity
multilingual embedding similarity
image perceptual-hash distance
identifier agreement
source-chain compatibility
edition/version indicators
rights or byline conflict
```

Suggested initial zones are deliberately conservative:

```
score >= 0.97:
    automatic candidate labelled HIGH, but only deterministic exact relations
    may auto-confirm

0.80 <= score < 0.97:
    clerical review

0.20 <= score < 0.80:
    retain candidate if relation is analytically important; otherwise sample
    for validation

score < 0.20:
    non-match unless a deterministic identifier conflicts
```

Thresholds must be learned or adjusted from the pilot gold standard. A single global threshold is inappropriate because translation detection, news syndication, person linkage and image matching have different error costs.

Human adjudication interface

The review interface should display:

original and normalised titles
work/expression/manifestation hierarchy
source URLs and identifiers
publication and retrieval times
side-by-side text with highlighted differences
image thumbnails and perceptual-hash score
named entities and quotations
rights statements
candidate relation and feature contributions
existing provenance chain
reviewer decision, confidence and rationale

Disagreements should be resolved by a third reviewer or consensus conference. The final relation must retain both initial decisions and the adjudication outcome.

Gold standard and validation design

Create a hand-labelled, stratified gold standard before fixing model thresholds. The sample should deliberately include easy and difficult cases:

Stratum	Minimum pilot emphasis
Exact binary and exact text duplicates	High enough to verify deterministic rules
Syndicated and mirrored news	Publishers and countries with differing editorial practices
Same text with different headlines	At least two languages or media systems
Revised articles	Multiple captures of live pages
Translations	Human and machine translations, including low-resource languages
Derivative visual works	Warming-stripe adaptations, posters, memes or remixes
Non-matches with similar titles	Prevent false merges
Same event, different artefacts	Prevent event-based over-merging
Same survey construct, different wording	Test harmonisation restraint
Historical images with visual similarity	Test false visual merges

Measure:

Metric	Definition and use
Precision	Confirmed correct positive decisions divided by all positive decisions

Metric	Definition and use
Recall	Correctly identified positive relations divided by all true positive relations in the gold sample
False merge rate	Distinct entities incorrectly combined or asserted as same
False split rate	Same entities or expressions left unlinked
Provenance coverage	Proportion with source, retrieval, checksum and acquisition activity
Rights coverage	Proportion with an explicit rights and ethics decision
Field completeness	Required populated fields by entity and source type
Unresolved rate	Proportion in <code>NEEDS_REVIEW</code> , disputed or missing-decision states
Source completeness	Coverage of the declared sampling frame, not merely record completeness
Reproducibility rate	Proportion of derived outputs rebuildable from retained inputs and code

False merges should receive greater weight than false splits for people, heritage objects, survey questions and rights-governed items because merging can destroy distinctions and propagate incorrect rights or identity claims.

Inter-coder agreement

Use Cohen's kappa only for two coders assigning nominal categories under a simple complete design. Use Krippendorff's alpha when there are more than two coders, missing ratings, varying numbers of coders, or ordinal and interval data. Krippendorff's alpha was designed as a general reliability coefficient across observers and measurement levels, whereas kappa is best understood as a two-rater, chance-corrected measure for categorical assignments and can behave problematically under severe category imbalance. ¹⁷

Report raw agreement, category prevalence and confusion matrices alongside kappa or alpha. Reliability is not validity: coders can agree while applying a concept incorrectly.

Separate validation exercises are required for:

Classification	Validation method
Relevance	Stratified sample of included and excluded records
Direct versus proxy affect	Double coding with original-language evidence
Heritage status	Independent rubric coding plus external-claim verification
Duplicate relationship	Pairwise gold standard and cluster-level inspection
Geographical assignment	Gazetteer and source-context verification
Language and translation	Native or expert review by language stratum
Public-release eligibility	Rights/ethics audit with deliberate high-risk cases

Construct validity should examine whether the affect and heritage categories behave as theorised without collapsing into media volume, recency, source prestige or English-language availability.

Rights, ethics and public-release governance

Governing principle

“Publicly accessible” does not mean that content may be scraped, analysed, copied or republished without limits.

Access on a website does not automatically resolve copyright, contractual terms, contextual privacy, platform expectations, consent, vulnerability, community authority or the ethical effects of redistributing content. AoIR’s Internet Research Ethics guidance emphasises context-sensitive ethical deliberation rather than treating online availability as blanket consent. Contextual-integrity theory similarly evaluates whether information flows remain appropriate to the social context in which the information was produced. ¹⁸

Rights and ethics decision fields

```
copyright_status
copyright_jurisdiction
rights_holder
licence_uri
rights_statement_uri
terms_of_service_review
research_exception_basis
personal_data_class
identifiability
sensitive_category
minor_involved
vulnerable_group
trauma_or_grief_context
deleted_post_status
consent_status
withdrawal_route
indigenous_data_flag
community_authority
community_protocol
local_contexts_notice_or_label
sensitive_location
internal_use_permission
public_metadata_permission
public_excerpt_permission
public_thumbnail_permission
public_full_text_permission
public_full_media_permission
embargo_end
takedown_contact
decision_date
```

decision_expiry

reviewer

RightsStatements.org supplies standardised statements for online cultural-heritage objects, but these are communications about assessed copyright status rather than licences. Local Contexts provides Traditional Knowledge and Biocultural Labels and Notices for community-specific authority, attribution, access and use protocols. ¹⁹

Public-release matrix

Material condition	Internal research	Public metadata	Excerpt/ thumbnail	Full content
Public domain or compatible open licence	Yes	Yes	Yes	Yes, subject to ethics and attribution
Copyrighted news article acquired lawfully	Often yes under institutional conditions	Usually yes	Only after jurisdiction-specific review	Normally no
Licensed survey microdata	Under licence	Study-level metadata usually	Questionnaire excerpts according to terms	Only where licence permits
Deleted social-media post	Restricted review	Usually de-identified metadata only	Normally no	No
Identifiable minor	Exceptional, approved research only	Aggregated or suppressed	No	No
Disaster survivor testimony	Consent- and context-dependent	Possibly generalised metadata	Only with explicit approval	Rarely
Oral history with consent	As consent permits	As consent permits	As consent permits	Only within consent and withdrawal terms
Indigenous/ community-controlled content	Only under community authority	Community-determined	Community-determined	Community-determined
Sensitive location	Yes with access controls	Generalised geography	Usually no precise coordinates	No precise coordinates

Material condition	Internal research	Public metadata	Excerpt/ thumbnail	Full content
Unknown copyright	Internal metadata and limited research use after review	Metadata-only	No until assessed	No
Platform content under restrictive terms	Only after legal/ ethics review	Limited metadata	No by default	No
Research-generated classification	Yes	Usually yes if source exposure is safe	Supporting evidence may be restricted	Underlying source rights govern

In Australia, fair dealing is purpose-specific and does not amount to a general quotation or open-republication right. Research and study, criticism or review, parody or satire, news reporting and specified other purposes may be covered, but fairness depends on context and amount. A thesis database should therefore store the legal basis for internal analysis separately from permission to publish the underlying material. ²⁰

Contextual privacy and social media

For social-media records, assess:

public, semi-public or closed setting
account visibility at collection
expected audience
searchability
user vulnerability
identifiability from quotations
sensitivity of topic
whether the post was deleted
whether the account was deleted
whether redistribution increases exposure
whether direct quotation enables reverse search

Deleted content should be marked immediately when deletion is discovered. Deletion can reasonably indicate a changed expectation of visibility and should trigger review, suppression from public outputs and, where feasible, withdrawal from analysis. Existing guidance on social-media research specifically identifies deleted posts and withdrawal as ethically significant rather than treating prior capture as unrestricted permission. ²¹

Direct quotations from ordinary users should be avoided or paraphrased where reverse-search identification is likely, unless consent, public-interest justification and ethics approval support quotation.

Indigenous data sovereignty and community control

For Aboriginal and Torres Strait Islander data in an Australian thesis, CARE should be applied alongside the AIATSIS Code of Ethics. The AIATSIS framework centres Indigenous self-determination, Indigenous leadership, impact and value, and sustainability and accountability; it applies to research involving Indigenous collections as well as direct research participation. ²²

Operational requirements include:

- identify relevant community authority
- record who may make access decisions
- do not infer community permission from institutional possession
- support community-specific access categories
- record attribution and terminology preferences
- support time-, gender-, ceremony- or season-based restrictions
- allow community correction and withdrawal
- avoid exposing restricted knowledge through metadata
- document benefit and reciprocity
- separate legal copyright from cultural authority

The database must permit community authority to override the system's default public-release logic. A technically FAIR object may appropriately remain inaccessible or available only through negotiated access.

Trauma, grief, minors and vulnerable communities

Records involving heat deaths, disaster survivors, displacement, grief, injury or loss should receive a harm review even when the source is institutional or already published. The review should consider dignity, sensationalism, re-identification, family expectations, cultural practices around death, and whether later reuse changes the original context.

For oral histories, consent should be granular:

- recording
- transcription
- translation
- research analysis
- quotation
- online publication
- deposit in archive
- future reuse
- name disclosure
- geographical disclosure
- withdrawal deadline or ongoing withdrawal process

A consent form is not sufficient if the intended new use falls outside what participants could reasonably understand at collection.

Takedown, correction and retraction

The public interface should publish a single contact route and process:

1. Acknowledge the request and create a restricted case record.
2. Temporarily suppress high-risk public content where plausible harm is alleged.
3. Preserve the internal evidential and audit record.
4. Notify the responsible rights/ethics reviewer.
5. Consult the claimant, rights holder or relevant community authority.
6. Decide removal, redaction, correction, retention or referral.
7. Record the basis, decision date and review date.
8. Republish corrected exports with a new version.
9. Retain a non-exposing tombstone when scholarly integrity requires it.

Corrections must not silently rewrite prior releases. A retracted public record should point to a safe explanation and replacement identifier where one exists.

Human-only decisions

The following must not be delegated to an automated classifier:

- whether an affect expression is direct fear rather than quotation, negation, sarcasm or accusation;
- whether a community has recognised an object as heritage;
- whether institutional recognition is equivalent to heritage status;
- whether two people, historical objects or contested places are the same;
- whether a translation is semantically equivalent for affect coding;
- whether culturally restricted material may be displayed;
- whether a deleted post remains ethically usable;
- whether a quotation presents unacceptable re-identification risk;
- whether a person is a minor when evidence is uncertain;
- whether public-interest justification outweighs foreseeable harm;
- whether a rights exception permits public display;
- whether Indigenous or community authority has been adequately established;
- whether conflicting heritage claims can be reconciled;
- whether a retraction or takedown should be granted.

Automated systems can identify missing fields, nominate candidates and calculate risk indicators. They cannot confer cultural authority or legal permission.

Systematic bias risks

The database can reproduce systematic bias through:

Source of bias	Likely consequence	Mitigation
English-dominant archives	English discourse appears more prevalent or historically important	Language quotas, local archive searches, translated discovery terms
Digitisation inequality	Well-funded countries appear to have richer heritage histories	Report source coverage and archive absence separately

Source of bias	Likely consequence	Mitigation
Newswire availability	Syndicated Western reporting dominates circulation networks	Preserve source lineage and sample local outlets
Keyword discovery	Explicit Western affect vocabulary is overrepresented	Concept- and context-based multilingual search
Machine translation	Affect intensity, politeness, irony and agency shift	Original-language adjudication
Multilingual embeddings	Low-resource languages generate fewer candidate matches	Per-language validation and lower reliance on model scores
Platform APIs	Current, highly engaged content is favoured over deleted or older material	Archive-based and non-platform sampling
Institutional heritage evidence	Formal museums outweigh community memory practices	Separate institutional and community recognition
Survey availability	Countries with repeated professional polling dominate opinion trends	Coverage matrices and no-data categories
Geocoding	Contemporary state boundaries overwrite historical or Indigenous geography	Time-bounded and multi-authority place records
Rights clearance	Openly licensed material appears more culturally important because it is easier to display	Separate research inclusion from public availability
Survivorship	Preserved objects appear more socially influential than lost objects	Record archival absence and source-chain limitations
Researcher coding	Thesis concepts are imposed on communities	Reflexive notes, community review and contestable assertions

Technical implementation, pilot corpus and final decision

Architecture comparison

Architecture	Strengths	Weaknesses	Thesis decision
PostgreSQL relational	Strong constraints, transactions, auditability, SQL analysis, mature backup	Verbose for arbitrary graph traversal	Use as authoritative core
PostgreSQL plus JSONB	Supports source-specific payloads and evolving metadata while retaining relational keys	JSON can become an ungoverned dumping ground	Use only for variable extensions and captured API metadata

Architecture	Strengths	Weaknesses	Thesis decision
Graph database	Natural exploration of dense reuse and circulation networks	Extra operational system; weaker fit for survey tables and strict validation	Do not use initially; export edge tables if needed
RDF triple store	Standards interoperability and semantic queries	High modelling and ontology-maintenance cost	JSON-LD/RDF export only
Hybrid relational/graph	Strongest long-term flexibility	Highest complexity, duplicated indexes and synchronisation risk	Consider after pilot only if graph queries become demonstrably limiting

Logical schema and indexing priorities

The logical design should group tables by schema:

```

core
  research_object, agent, place, concept

climate
  climate_event, climate_observation, institutional_milestone

artifact
  artifact_work, artifact_expression, artifact_manifestation,
  archived_item, circulation_event, reuse_relation

survey
  public_opinion_study, survey_question, survey_observation

interpretation
  affect_assertion, heritage_claim, heritage_assessment

governance
  provenance_event, provenance_used, provenance_generated,
  rights_statement, review_decision, gate_assessment,
  status_transition, source_locator

release
  publication_projection, release_manifest

```

Recommended indexes are:

```

B-tree:
  dates, status, object type, foreign keys, country codes,
  study/wave, question, gate result

```

Unique or partial unique:
 checksum + algorithm for bitstream identity
 external identifier within identifier scheme
 versioned local IDs

GIN:
 JSONB payloads
 full-text vectors
 arrays of controlled concepts where denormalised for search

GiST or GIN with pg_trgm:
 titles, agent aliases, source names

Spatial:
 place geometry and event footprints using PostGIS

Composite:
 survey_question_id + place_id + field period + response category
 object_id + gate_code + active decision
 relation source + target + type

PostgreSQL documents GIN indexing for JSONB and full-text search, while `pg_trgm` provides indexed trigram similarity. These are appropriate for candidate search but should not replace relational constraints or reviewed entity links. ²³

Sample PostgreSQL DDL

The following is representative rather than a complete production migration.

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
CREATE EXTENSION IF NOT EXISTS pg_trgm;
CREATE EXTENSION IF NOT EXISTS unaccent;
-- Optional when spatial analysis is required:
-- CREATE EXTENSION IF NOT EXISTS postgis;

CREATE TYPE object_status AS ENUM (
  'REFERENCE_ONLY',
  'SOURCE_LEAD',
  'CANDIDATE_RECORD',
  'PROVENANCE_VERIFIED',
  'VERIFIED_CULTURAL_RECORD',
  'HERITAGE_CANDIDATE',
  'COMMUNITY_RECOGNISED_HERITAGE',
  'INSTITUTIONALLY_RECOGNISED_HERITAGE',
  'RESEARCH_ONLY',
  'PUBLIC_RELEASE_APPROVED',
  'NEEDS_REVIEW',
  'QUARANTINED',
  'EXCLUDED',
  'RETRACTED'
```

```

);

CREATE TYPE review_result AS ENUM (
    'PASS', 'FAIL', 'REVIEW', 'NOT_APPLICABLE', 'SUPERSEDED'
);

CREATE TYPE release_level AS ENUM (
    'METADATA_ONLY',
    'THUMBNAIL',
    'EXCERPT',
    'FULL_TEXT',
    'FULL_MEDIA',
    'RESTRICTED',
    'EMBARGOED',
    'NO_PUBLIC_RELEASE'
);

CREATE TABLE research_object (
    object_id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    object_type        text NOT NULL,
    current_status     object_status NOT NULL DEFAULT 'SOURCE_LEAD',
    schema_version     text NOT NULL,
    created_at         timestampz NOT NULL DEFAULT now(),
    retired_at         timestampz,
    CHECK (retired_at IS NULL OR retired_at >= created_at)
);

CREATE TABLE agent (
    agent_id           uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
    agent_type         text NOT NULL,
    canonical_name     text NOT NULL,
    aliases            jsonb NOT NULL DEFAULT '[]'::jsonb,
    active_start       date,
    active_end         date,
    identifiers        jsonb NOT NULL DEFAULT '{}'::jsonb,
    identity_certainty numeric(4,3),
    notes             text
);

CREATE INDEX agent_name_trgm_idx
    ON agent USING gin (canonical_name gin_trgm_ops);

CREATE TABLE place (
    place_id           uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
    preferred_name     text NOT NULL,
    place_type        text NOT NULL,
    country_code       char(2),
    historical_names   jsonb NOT NULL DEFAULT '[]'::jsonb,
    external_identifiers jsonb NOT NULL DEFAULT '{}'::jsonb,

```

```

        latitude          numeric(9,6),
        longitude         numeric(9,6),
        spatial_uncertainty_m numeric,
        sensitive_location boolean NOT NULL DEFAULT false,
        public_location_label text
    );

CREATE INDEX place_name_trgm_idx
    ON place USING gin (preferred_name gin_trgm_ops);

CREATE TABLE concept_scheme (
    scheme_id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    scheme_code        text NOT NULL,
    scheme_version     text NOT NULL,
    title              text NOT NULL,
    issued_at          timestampz NOT NULL,
    UNIQUE (scheme_code, scheme_version)
);

CREATE TABLE concept (
    concept_id         uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    scheme_id          uuid NOT NULL REFERENCES concept_scheme(scheme_id),
    concept_code       text NOT NULL,
    preferred_label    text NOT NULL,
    definition         text NOT NULL,
    broader_id         uuid REFERENCES concept(concept_id),
    deprecated_at      timestampz,
    UNIQUE (scheme_id, concept_code)
);

CREATE TABLE concept_label (
    concept_id         uuid NOT NULL REFERENCES concept(concept_id),
    language_tag       text NOT NULL,
    label_type         text NOT NULL,
    label              text NOT NULL,
    source_note        text,
    PRIMARY KEY (concept_id, language_tag, label_type, label)
);

CREATE TABLE climate_event (
    climate_event_id   uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
    event_type         text NOT NULL,
    preferred_name     text NOT NULL,
    event_start        timestampz,
    event_end          timestampz,
    date_lexical       text,
    date_certainty     text NOT NULL,
    scope_note         text NOT NULL,
    CHECK (
        event_end IS NULL OR event_start IS NULL
    )
);

```

```

        OR event_end >= event_start
    )
);

CREATE TABLE climate_event_place (
    climate_event_id uuid NOT NULL
                        REFERENCES climate_event(climate_event_id),
    place_id          uuid NOT NULL REFERENCES place(place_id),
    place_role        text NOT NULL,
    evidence_note     text,
    PRIMARY KEY (climate_event_id, place_id, place_role)
);

CREATE TABLE climate_observation (
    climate_observation_id uuid PRIMARY KEY
                            REFERENCES research_object(object_id),
    climate_event_id       uuid REFERENCES climate_event(climate_event_id),
    place_id               uuid REFERENCES place(place_id),
    variable_code          text NOT NULL,
    measurement_type       text NOT NULL,
    statistic_type         text NOT NULL,
    value_original         numeric NOT NULL,
    unit_original          text NOT NULL,
    value_celsius          numeric,
    period_start           timestampz NOT NULL,
    period_end             timestampz,
    anomaly_baseline_start smallint,
    anomaly_baseline_end   smallint,
    dataset_name           text NOT NULL,
    dataset_version        text NOT NULL,
    uncertainty_value       numeric,
    uncertainty_type        text,
    source_item_id         uuid,
    CHECK (
        period_end IS NULL OR period_end >= period_start
    )
);

CREATE INDEX climate_obs_time_place_idx
    ON climate_observation(place_id, period_start);

CREATE TABLE institutional_milestone (
    milestone_id          uuid PRIMARY KEY
                            REFERENCES research_object(object_id),
    milestone_type        text NOT NULL,
    title                 text NOT NULL,
    responsible_agent_id  uuid REFERENCES agent(agent_id),
    event_date_start      date,
    event_date_end        date,
    stable_identifier     text,
    description            text NOT NULL
);

```

```

);

CREATE TABLE artifact_work (
    work_id          uuid PRIMARY KEY
                     REFERENCES research_object(object_id),
    work_type        text NOT NULL,
    preferred_title   text NOT NULL,
    creation_start    date,
    creation_end      date,
    abstract          text,
    creator_uncertain boolean NOT NULL DEFAULT false
);

CREATE INDEX artifact_work_title_trgm_idx
ON artifact_work USING gin (preferred_title gin_trgm_ops);

CREATE TABLE artifact_expression (
    expression_id     uuid PRIMARY KEY
                     REFERENCES research_object(object_id),
    work_id           uuid NOT NULL REFERENCES artifact_work(work_id),
    expression_type    text NOT NULL,
    language_tag       text NOT NULL,
    script_code        text,
    revision_label     text,
    source_expression_id uuid REFERENCES
artifact_expression(expression_id),
    translation_method text,
    translator_agent_id uuid REFERENCES agent(agent_id),
    semantic_notes     text
);

CREATE INDEX artifact_expression_work_idx
ON artifact_expression(work_id);

CREATE TABLE artifact_manifestation (
    manifestation_id  uuid PRIMARY KEY
                     REFERENCES research_object(object_id),
    expression_id     uuid NOT NULL
                     REFERENCES artifact_expression(expression_id),
    publisher_agent_id uuid REFERENCES agent(agent_id),
    publication_start timestampz,
    publication_end   timestampz,
    channel_type       text NOT NULL,
    edition_statement text,
    manifestation_title text,
    canonical_url      text,
    source_identifier  text
);

CREATE INDEX manifestation_expression_idx
ON artifact_manifestation(expression_id);

```

```

CREATE INDEX manifestation_url_idx
  ON artifact_manifestation(canonical_url);

CREATE TABLE archived_item (
  archived_item_id      uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
  manifestation_id      uuid REFERENCES
                        artifact_manifestation(manifestation_id),
  storage_uri           text NOT NULL,
  original_filename     text,
  mime_type             text NOT NULL,
  byte_size             bigint NOT NULL CHECK (byte_size >= 0),
  checksum_algorithm    text NOT NULL DEFAULT 'SHA-256',
  checksum_value        text NOT NULL,
  retrieval_timestamp   timestampz NOT NULL,
  format_valid          boolean,
  original_copy         boolean NOT NULL DEFAULT true,
  access_class          text NOT NULL DEFAULT 'RESEARCH_ONLY',
  UNIQUE (checksum_algorithm, checksum_value, storage_uri)
);

CREATE INDEX archived_checksum_idx
  ON archived_item(checksum_algorithm, checksum_value);

CREATE TABLE public_opinion_study (
  study_id              uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
  programme_name        text NOT NULL,
  wave_label            text NOT NULL,
  sponsor_agent_id      uuid REFERENCES agent(agent_id),
  fieldwork_start       date,
  fieldwork_end         date,
  target_population     text NOT NULL,
  sample_design         text NOT NULL,
  mode                  text NOT NULL,
  weighting_documented  boolean NOT NULL,
  source_identifier     text
);

CREATE TABLE survey_question (
  survey_question_id    uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
  study_id              uuid NOT NULL
                        REFERENCES public_opinion_study(study_id),
  question_code         text,
  language_tag          text NOT NULL,
  exact_wording         text NOT NULL,
  introductory_text     text,
  question_order        numeric,
  response_options      jsonb NOT NULL,

```

```

        coding_direction      text,
        construct_code        text NOT NULL,
        translation_method     text,
        source_question_id     uuid
                                REFERENCES survey_question(survey_question_id)
    );

CREATE INDEX survey_question_text_trgm_idx
    ON survey_question USING gin (exact_wording gin_trgm_ops);

CREATE TABLE survey_observation (
    survey_observation_id  uuid PRIMARY KEY
                            REFERENCES research_object(object_id),
    survey_question_id     uuid NOT NULL
                            REFERENCES survey_question(survey_question_id),
    place_id               uuid NOT NULL REFERENCES place(place_id),
    subgroup_definition     jsonb NOT NULL DEFAULT '{}'::jsonb,
    response_code           text NOT NULL,
    estimate                numeric NOT NULL,
    estimate_unit           text NOT NULL,
    unweighted_n            integer,
    weighted_n              numeric,
    standard_error          numeric,
    confidence_low           numeric,
    confidence_high         numeric,
    weight_name             text,
    derivation_code_version text
);

CREATE INDEX survey_obs_question_place_idx
    ON survey_observation(survey_question_id, place_id);

CREATE TABLE affect_assertion (
    affect_assertion_id  uuid PRIMARY KEY
                        REFERENCES research_object(object_id),
    evidence_object_id   uuid NOT NULL
                        REFERENCES research_object(object_id),
    affect_concept_id    uuid NOT NULL REFERENCES concept(concept_id),
    experiencer_agent_id uuid REFERENCES agent(agent_id),
    target_concept_id    uuid REFERENCES concept(concept_id),
    evidence_text         text,
    evidence_locator      jsonb NOT NULL,
    coding_method         text NOT NULL,
    confidence            numeric(4,3) NOT NULL
                        CHECK (confidence BETWEEN 0 AND 1),
    rule_version          text NOT NULL,
    reviewer_agent_id    uuid REFERENCES agent(agent_id)
);

CREATE TABLE circulation_event (
    circulation_event_id  uuid PRIMARY KEY

```

```

        REFERENCES research_object(object_id),
circulated_object_id uuid NOT NULL
        REFERENCES research_object(object_id),
circulation_type      text NOT NULL,
event_start           timestamptz,
event_end             timestamptz,
channel_agent_id      uuid REFERENCES agent(agent_id),
place_id              uuid REFERENCES place(place_id),
audience_evidence    text,
evidence_object_id    uuid REFERENCES research_object(object_id)
);

CREATE TABLE reuse_relation (
    reuse_relation_id  uuid PRIMARY KEY
        REFERENCES research_object(object_id),
    source_object_id   uuid NOT NULL REFERENCES research_object(object_id),
    target_object_id   uuid NOT NULL REFERENCES research_object(object_id),
    relation_type      text NOT NULL,
    asserted_date      date,
    confidence          numeric(4,3) NOT NULL
        CHECK (confidence BETWEEN 0 AND 1),
    evidence_object_id uuid REFERENCES research_object(object_id),
    reviewer_agent_id  uuid REFERENCES agent(agent_id),
    CHECK (source_object_id <> target_object_id)
);

CREATE INDEX reuse_relation_endpoints_idx
    ON reuse_relation(source_object_id, target_object_id, relation_type);

CREATE TABLE heritage_claim (
    heritage_claim_id  uuid PRIMARY KEY
        REFERENCES research_object(object_id),
    claimed_object_id  uuid NOT NULL REFERENCES research_object(object_id),
    claimant_agent_id  uuid NOT NULL REFERENCES agent(agent_id),
    claim_type         text NOT NULL,
    claim_date         date,
    rationale          text NOT NULL,
    exact_evidence     text,
    evidence_object_id uuid REFERENCES research_object(object_id)
);

CREATE TABLE heritage_assessment (
    heritage_assessment_id uuid PRIMARY KEY
        REFERENCES research_object(object_id),
    assessed_object_id     uuid NOT NULL
        REFERENCES research_object(object_id),
    rubric_version         text NOT NULL,
    criterion_scores       jsonb NOT NULL,
    outcome                text NOT NULL,
    rationale              text NOT NULL,
    confidence             numeric(4,3)

```

```

        CHECK (confidence BETWEEN 0 AND 1),
assessor_agent_id    uuid NOT NULL REFERENCES agent(agent_id),
assessed_at          timestampz NOT NULL DEFAULT now(),
supersedes_id        uuid REFERENCES
                    heritage_assessment(heritage_assessment_id)
);

CREATE TABLE provenance_event (
    provenance_event_id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    event_type          text NOT NULL,
    started_at          timestampz NOT NULL,
    ended_at            timestampz,
    associated_agent_id uuid REFERENCES agent(agent_id),
    software_name        text,
    software_version     text,
    code_commit          text,
    parameters           jsonb NOT NULL DEFAULT '{} '::jsonb,
    outcome              text NOT NULL,
    CHECK (ended_at IS NULL OR ended_at >= started_at)
);

CREATE TABLE provenance_used (
    provenance_event_id uuid NOT NULL
                        REFERENCES provenance_event(provenance_event_id),
    object_id            uuid NOT NULL REFERENCES research_object(object_id),
    role                 text NOT NULL,
    PRIMARY KEY (provenance_event_id, object_id, role)
);

CREATE TABLE provenance_generated (
    provenance_event_id uuid NOT NULL
                        REFERENCES provenance_event(provenance_event_id),
    object_id            uuid NOT NULL REFERENCES research_object(object_id),
    role                 text NOT NULL,
    PRIMARY KEY (provenance_event_id, object_id, role)
);

CREATE TABLE source_locator (
    source_locator_id    uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    object_id            uuid NOT NULL REFERENCES research_object(object_id),
    locator_type          text NOT NULL,
    locator_value         text NOT NULL,
    observed_at           timestampz,
    canonicalised_value   text,
    redirect_chain        jsonb,
    UNIQUE (object_id, locator_type, locator_value)
);

CREATE TABLE rights_statement (
    rights_statement_id  uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    object_id            uuid NOT NULL REFERENCES research_object(object_id),

```

```

copyright_status    text NOT NULL,
rights_holder_id    uuid REFERENCES agent(agent_id),
licence_uri         text,
rights_statement_uri text,
privacy_class       text NOT NULL,
community_protocol  jsonb,
internal_use        boolean NOT NULL,
release_level       release_level NOT NULL,
embargo_end         date,
decision_basis      text NOT NULL,
reviewer_agent_id   uuid NOT NULL REFERENCES agent(agent_id),
decided_at          timestampz NOT NULL DEFAULT now(),
supersedes_id       uuid REFERENCES rights_statement(rights_statement_id)
);

```

```

CREATE TABLE review_decision (
  review_decision_id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  object_id          uuid NOT NULL REFERENCES research_object(object_id),
  decision_type      text NOT NULL,
  decision           text NOT NULL,
  reviewer_agent_id  uuid NOT NULL REFERENCES agent(agent_id),
  rule_version       text NOT NULL,
  evidence           jsonb NOT NULL,
  rationale          text NOT NULL,
  decided_at         timestampz NOT NULL DEFAULT now(),
  supersedes_id      uuid REFERENCES review_decision(review_decision_id)
);

```

```

CREATE TABLE gate_assessment (
  gate_assessment_id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  object_id          uuid NOT NULL REFERENCES research_object(object_id),
  gate_code          text NOT NULL,
  result            review_result NOT NULL,
  rule_version       text NOT NULL,
  automatic_checks   jsonb NOT NULL DEFAULT '{} '::jsonb,
  evidence           jsonb NOT NULL DEFAULT '{} '::jsonb,
  reviewer_agent_id  uuid REFERENCES agent(agent_id),
  reviewed_at        timestampz NOT NULL DEFAULT now(),
  decision_reason    text NOT NULL,
  supersedes_id      uuid REFERENCES gate_assessment(gate_assessment_id)
);

```

```

CREATE INDEX gate_review_queue_idx
  ON gate_assessment(gate_code, result, reviewed_at);

```

```

CREATE TABLE status_transition (
  transition_id      uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  object_id          uuid NOT NULL REFERENCES research_object(object_id),
  previous_status    object_status,
  new_status         object_status NOT NULL,
  authorised_by      uuid NOT NULL REFERENCES agent(agent_id),

```

```

rule_version      text NOT NULL,
rationale         text NOT NULL,
occurred_at       timestampz NOT NULL DEFAULT now(),
review_decision_id uuid REFERENCES review_decision(review_decision_id)
);

```

Private/public separation should be implemented through restricted schemas, roles, row-level security and public projection views rather than application conventions alone. PostgreSQL row-security policies can restrict `SELECT`, `INSERT`, `UPDATE` and `DELETE` operations by role and row condition. ²⁴

Sample JSON-LD research record

```

{
  "@context": {
    "dcterm": "http://purl.org/dc/terms/",
    "prov": "http://www.w3.org/ns/prov#",
    "schema": "https://schema.org/",
    "skos": "http://www.w3.org/2004/02/skos/core#",
    "ft": "https://example.org/fear-temperature/"
  },
  "@id": "ft:work/0187",
  "@type": ["schema:CreativeWork", "ft:ArtifactWork"],
  "dcterm:title": {
    "@value": "Heatwave public-information report",
    "@language": "en"
  },
  "ft:workType": "institutional_report",
  "ft:expression": [
    {
      "@id": "ft:expression/0241",
      "@type": "ft:ArtifactExpression",
      "dcterm:language": "en-AU",
      "ft:expressionType": "original",
      "ft:manifestation": {
        "@id": "ft:manifestation/0310",
        "@type": "ft:ArtifactManifestation",
        "dcterm:issued": "2020-03-17",
        "dcterm:publisher": {
          "@id": "ft:agent/bom",
          "@type": "schema:Organization"
        },
      },
      "ft:archivedItem": {
        "@id": "ft:item/0552",
        "@type": "ft:ArchivedItem",
        "schema:encodingFormat": "application/pdf",
        "ft:checksum": {
          "algorithm": "SHA-256",
          "value": "5f51...example"
        },
      },
      "prov:wasGeneratedBy": {

```

```

        "@id": "ft:prov/acquisition-1072",
        "@type": "prov:Activity",
        "prov:endedAtTime": "2026-07-14T03:21:18Z",
        "ft:codeCommit": "7a1c9d2"
      }
    }
  }
},
"ft:affectAssertion": {
  "@id": "ft:affect/0089",
  "@type": "ft:AffectAssertion",
  "ft:affectClass": {
    "@id": "ft:concept/perceived-threat",
    "skos:prefLabel": "perceived threat"
  },
  "ft:evidenceLocator": {
    "page": 4,
    "paragraph": 2
  },
  "ft:confidence": 0.91,
  "ft:ruleVersion": "affect-1.3",
  "prov:wasAttributedTo": {
    "@id": "ft:agent/reviewer-03"
  }
},
"dcterms:rights": {
  "@id": "ft:rights/0711",
  "ft:releaseLevel": "METADATA_ONLY",
  "ft:internalUse": true,
  "ft:publicFullText": false
},
"ft:status": "VERIFIED_CULTURAL_RECORD",
"ft:publicReleaseApproved": false
}

```

Sample duplicate and reuse relation

```

{
  "relation_id": "rel-44d2",
  "source_object_id": "expression-en-193",
  "target_object_id": "expression-fr-208",
  "relation_type": "TRANSLATION_OF",
  "candidate_generation": {
    "title_similarity": 0.41,
    "multilingual_embedding_similarity": 0.94,
    "shared_named_entities": 7,
    "publication_time_difference_hours": 3.2
  },
  "probabilistic_score": 0.92,

```

```

"automatic_status": "CLERICAL_REVIEW",
"human_decision": {
  "decision": "CONFIRMED",
  "reviewer": "agent-reviewer-07",
  "confidence": 0.97,
  "evidence": [
    "matching newswire attribution",
    "paragraph-level semantic correspondence",
    "same quoted sources"
  ],
  "rule_version": "linkage-news-1.2"
}
}

```

The English and French expressions remain separate. Their manifestations, archived files, circulation events, rights and later reuse can differ.

Sample survey record

```

{
  "study": {
    "study_id": "study-ess8",
    "programme": "European Social Survey",
    "wave": "Round 8",
    "fieldwork_start": "2016-08-01",
    "fieldwork_end": "2017-12-31",
    "mode": "face-to-face",
    "weighting_documented": true
  },
  "question": {
    "question_id": "question-clmchngr",
    "language": "en-GB",
    "exact_wording": "How worried are you about climate change?",
    "response_options": [
      {"code": 1, "label": "Not at all worried"},
      {"code": 2, "label": "Not very worried"},
      {"code": 3, "label": "Somewhat worried"},
      {"code": 4, "label": "Very worried"},
      {"code": 5, "label": "Extremely worried"}
    ],
    "construct": "DIRECT_WORRY_ANXIETY",
    "fear_classification": "NOT_DIRECT_FEAR",
    "question_order_preserved": true
  },
  "observation": {
    "place": "FR",
    "response_code": 4,
    "estimate": 0.27,
    "estimate_unit": "weighted_proportion",
    "unweighted_n": 2044,

```

```

    "weight_name": "analysis_weight",
    "derived": true,
    "derivation_code_version": "survey-harmonisation-0.4.0"
  }
}

```

Sample heritage assessment

```

{
  "assessment_id": "ha-0034",
  "assessed_object_id": "work-warming-stripes",
  "rubric_version": "heritage-1.1",
  "criterion_scores": {
    "deliberate_preservation": 2,
    "temporal_persistence": 2,
    "value_attribution": 1,
    "collective_memory": 1,
    "transmission": 2,
    "transformative_reuse": 2
  },
  "outcome": "HERITAGE_CANDIDATE",
  "recognition_status": "NO_FORMAL_RECOGNITION_ASSERTED",
  "evidence": [
    "preserved institutional and university collection instances",
    "repeated educational reuse",
    "documented local adaptations",
    "cross-platform remixes and public displays"
  ],
  "confidence": 0.84,
  "reviewers": ["reviewer-heritage-01", "reviewer-heritage-02"],
  "unresolved_issue": "Whether broad reuse demonstrates durable collective-memory status",
  "public_label": "Research-assessed heritage candidate"
}

```

The warming-stripes graphic is particularly useful as a disputed pilot case because it links scientific temperature data to a highly reusable visual expression. The original graphic presents annual temperatures as sequential colour stripes, and later institutional and educational uses provide evidence with which to test manifestation, circulation and reuse relations. Those facts support investigation; they do not automatically prove heritage status. ²⁵

Pilot corpus plan

The pilot should be a **purposeful stress test**, not an attempt to collect everything concerning climate fear.

Proposed case clusters

Cluster	Rationale	Languages	Key evidence types
European heatwave of 2003 and retrospective remembrance	Mature physical, mortality, policy, media and retrospective evidence; supports heritage persistence testing	French, English, German	Station/gridded climate data, WMO reports, newspapers, surveys, anniversaries, exhibitions or teaching reuse
Australian extreme heat and 2019–20 crisis period	Strong official climate documentation and clear rights/ethics issues around disaster testimony and imagery	English; selected community-governed Indigenous metadata only with authority	Bureau of Meteorology data, government statements, media, activist visuals, oral-history or museum metadata
India–Pakistan heatwave of 2022	Cross-border and multilingual event; tests Hindi, Urdu and English linkage, local versus international framing and uneven archive availability	Hindi, Urdu, English	National and international reports, local media, public statements, visual communication
Warming-stripes visual lineage from 2018 onward	Tests data-to-artifact transformation, translation without words, remix, display, education, preservation and contested heritage assessment	Language-light visual object plus multilingual captions	Original data visualisation, institutional reproductions, campaign use, teaching and adaptation

The 2003 European heatwave was a major multi-country extreme-heat event and has accumulated substantial retrospective health and historical documentation. Australia's 2019 climate statement identifies 2019 as the country's warmest year in the national dataset then available. The 2022 India–Pakistan event provides a major transnational heat case with official international reporting and attribution research. ²⁶

Target pilot size

Object	Target
Climate events	12–20
Climate datasets or source editions	6–10
Climate observations	3,000–8,000 structured rows
Institutional milestones	30–50
Artifact works	220–320
Artifact expressions	320–480

Object	Target
Artifact manifestations	450–700
Archived items	400–600
Public-opinion studies or waves	6–10
Survey questions	30–60
Survey observations	250–600
Affect assertions	300–500
Circulation events	180–300
Reuse relations	150–250
Heritage claims	25–50
Heritage assessments	40–70
Rights statements	One current statement for every archived or releasable object
Deliberate linkage test pairs	500–1,000 candidate pairs

The pilot should deliberately include at least:

50 exact binary duplicates
50 exact text duplicates with different acquisition contexts
50 syndicated manifestations
30 revised articles
40 translations
30 excerpts or quotations
25 visual derivatives or remixes
20 same-event/non-duplicate pairs
20 rights-restricted items
10 deleted or withdrawn-content scenarios
15 disputed heritage assessments

The climate component must include at least two physically different data forms: for example, station or daily maximum observations and gridded, reanalysis or anomaly data. Official documentation demonstrates why this distinction matters: heatwave definitions and impacts depend on local climatic conditions, while annual national anomalies use different spatial and temporal aggregation. ²⁷

Pilot success criteria

The pilot is successful only if it demonstrates that:

- every analytical result can be traced to raw bytes, source, retrieval event, checksum, code and rule version;
- work, expression, manifestation and item distinctions resolve actual duplicate and translation cases;

- no direct-fear result lacks a quotable, locatable or questionnaire-based evidential basis;
- heritage candidates can be separated from cultural records and formal recognition;
- restricted objects remain analytically usable without leaking into public exports;
- at least three languages can be searched and reviewed;
- duplicate models meet relation-specific precision targets;
- rights and provenance coverage exceed pre-registered minimums;
- the public export can be regenerated deterministically.

Recommended pilot thresholds are:

```
provenance coverage >= 98%
rights-decision coverage = 100% for archived items
G0 technical pass >= 97%
public-release leakage incidents = 0
false merge rate for reviewed identity relations < 1%
precision for auto-confirmed exact duplicates >= 99.5%
precision for confirmed translation/syndication relations >= 95%
unresolved mandatory gate decisions < 5%
double-coded affect alpha >= 0.80 where sample size permits
heritage assessment alpha >= 0.67 initially, followed by rubric revision
```

Thresholds should be interpreted with confidence intervals and category prevalence, not as universal constants.

Public interface requirements

The public interface should expose only release projections and should clearly label:

```
source record
research interpretation
heritage candidate versus recognised heritage
language and translation method
rights statement
available media level
provenance summary
record and rule version
correction or takedown route
```

Users should be able to search by event, place, language, date, affect evidence, artefact type, circulation relation and heritage status. A public visualisation may show aggregate networks, but restricted nodes should be suppressed or represented only by non-identifying counts.

No public API should return private fields merely because the caller knows an object UUID.

Backup and preservation

Use:

- nightly encrypted database dumps
- regular tested restores
- versioned object-storage replication
- off-site checksum manifests
- periodic fixity verification
- release-level immutable snapshots
- Git repository mirrors
- dependency and container snapshots
- access-log retention appropriate to ethics policy

PostgreSQL documents SQL dumps, filesystem backups and continuous archiving as distinct backup approaches; point-in-time recovery requires base backups and archived write-ahead logs rather than relying only on `pg_dump`. ²⁸

A backup is not valid until restoration is tested. Restricted data backups must be encrypted, access-controlled and subject to the same retention and takedown policy as the primary system.

Risk register

Risk	Probability	Impact	Mitigation
Earlier Deep Research taxonomies conflict with this model	High until supplied	High	Treat all vocabularies and thresholds as provisional; create crosswalk and migration
Scope becomes too large	High	High	Freeze pilot cases, languages and target counts before ingestion
Work/expression distinctions are inconsistently applied	Medium	High	Decision guide, examples and double review
Rights status remains unknown	High	High	Metadata-only default and mandatory G6
Public/private leakage	Low-medium	Critical	Separate schemas, roles, row security, release tests
False merges erase circulation evidence	Medium	High	Link-not-delete principle and conservative thresholds
Excessive false splits	Medium	Medium	Candidate-generation evaluation and cluster review
English-language dominance	High	High	Language quotas and per-language coverage reporting
Machine translation changes affect meaning	High	High	Original-language evidence and human adjudication
Heritage inflation	Medium	High	Separate claims, assessments and recognition statuses
Institutional bias in heritage evidence	High	High	Community recognition as separate pathway

Risk	Probability	Impact	Mitigation
Social-media re-identification	Medium	High	Paraphrase, suppress handles, metadata-only publication
Deleted posts remain in public exports	Low-medium	High	Periodic availability check and withdrawal workflow
Dataset versions change	High	Medium	Pin editions, checksums and variable definitions
Source websites disappear	High	High	Lawful preservation captures and stable archive IDs
Review burden exceeds thesis capacity	High	High	Risk-based review and reduced corpus
Graph analysis becomes slow	Low at pilot scale	Medium	Indexed edge tables and materialized views
Vocabulary drift breaks reproducibility	Medium	Medium	Immutable scheme versions
Community restrictions are reduced to generic copyright	Medium	Critical	CARE/AIATSIS governance and Local Contexts-compatible fields
Legal jurisdiction is unclear	Medium	High	Record jurisdiction and obtain institutional advice
Researcher's own heritage theory dominates evidence	Medium	High	Reflexive notes, external claims and disputed status

Estimated labour and technical complexity

A realistic thesis-scale implementation for the proposed pilot is approximately:

Work package	Person-hours
Requirements reconciliation with Deep Research One and Two	30–50
Schema, vocabulary and rule design	70–100
Database migrations and ingestion framework	80–130
File acquisition and preservation workflow	60–100
Entity and source metadata creation	120–180
Survey harmonisation	50–90
Multilingual review and translation documentation	70–130
Affect and heritage coding	120–190
Deduplication model and adjudication	80–130
Rights and ethics review	80–140

Work package	Person-hours
QA, gold standard and validation	70–110
Public release and documentation	60–100
Total	890–1,450

A solo Master thesis should therefore use the lower end of the proposed corpus, reuse authoritative metadata where legally and methodologically appropriate, and avoid building a bespoke graph database, ontology reasoner or large public-facing web platform. The public deliverable can be a static or modest server-rendered catalogue backed by versioned exports.

Final GO / MODIFY / NO-GO decision

Decision: MODIFY, then GO for a bounded pilot.

The project is methodologically viable if the following modifications are made before data-scale expansion:

1. Insert and reconcile the missing Deep Research One and Two materials.
2. Freeze a historical and geographical scope.
3. Approve version one of the affect and heritage vocabularies.
4. Establish an ethics and rights escalation route.
5. Confirm whether any Indigenous or community-controlled material is within scope and, if so, identify the relevant authority before acquisition.
6. Limit the pilot to approximately 400–600 archived items.
7. Implement PostgreSQL and immutable object storage before considering graph infrastructure.
8. Create the gold-standard sample before fixing linkage thresholds.
9. Default unknown-rights material to internal research and metadata-only publication.
10. Require human approval for direct fear, heritage recognition, contested identity, culturally restricted access and public release.

The project becomes **NO-GO** for public release if rights and ethics decisions cannot be completed, if provenance cannot be traced to original or trustworthy source chains, or if community-controlled material is included without authority. It becomes **NO-GO** as a claim about “fear” if direct affect cannot be distinguished from concern, threat, rhetoric, attention and behaviour proxies. It becomes **NO-GO** as a heritage database if researcher selection, virality or archive capture is presented as heritage recognition.

The strongest defensible thesis contribution is not a claim to have constructed a universal archive of climate fear. It is a reproducible method for showing **how scientific temperature evidence, public affect, cultural expression, circulation, preservation and heritage-making become related without being conflated.**

¹ ²³ F.35. pg_trgm — support for similarity of text using trigram ...
https://www.postgresql.org/docs/current/pgtrgm.html?utm_source=chatgpt.com

² ¹⁰ PROV-O: The PROV Ontology
https://www.w3.org/TR/prov-o/?utm_source=chatgpt.com

- 3 **DCMI Metadata Terms**
https://dublincore.org/specifications/dublin-core/dcmi-terms/2020-01-20/?utm_source=chatgpt.com
- 4 **The FAIR Guiding Principles for scientific data ...**
https://www.nature.com/articles/sdata201618?utm_source=chatgpt.com
- 5 **SKOS Simple Knowledge Organization System Reference**
https://www.w3.org/TR/skos-reference/?utm_source=chatgpt.com
- 6 **FULL GUIDELINES**
https://ccsg.isr.umich.edu/wp-content/uploads/2019/06/CCSG_Full_Guidelines_2016_Version.pdf?utm_source=chatgpt.com
- 7 **ESS8_climate_final_module_te...**
https://www.europeansocialsurvey.org/sites/default/files/2023-06/ESS8_climate_final_module_template.pdf?utm_source=chatgpt.com
- 8 **RFC 5646 - Tags for Identifying Languages**
https://datatracker.ietf.org/doc/html/rfc5646?utm_source=chatgpt.com
- 9 **Language-agnostic BERT Sentence Embedding**
https://aclanthology.org/2022.acl-long.62/?utm_source=chatgpt.com
- 11 **Extended Date/Time Format (EDTF) Specification**
https://www.loc.gov/standards/datetime/?utm_source=chatgpt.com
- 12 **GeoNames Ontology - Geo Semantic Web**
https://www.geonames.org/ontology?utm_source=chatgpt.com
- 13 **The Unified Code for Units of Measure**
https://ucum.org/ucum?utm_source=chatgpt.com
- 14 **Hash Functions | CSRC**
https://csrc.nist.gov/projects/hash-functions?utm_source=chatgpt.com
- 15 **Recommended Formats Statement – table of contents**
https://www.loc.gov/preservation/resources/rfs/TOC.html?utm_source=chatgpt.com
- 16 **A Theory for Record Linkage**
https://www.tandfonline.com/doi/abs/10.1080/01621459.1969.10501049?utm_source=chatgpt.com
- 17 **Computing Krippendorff's Alpha-Reliability**
https://www.asc.upenn.edu/sites/default/files/2021-03/Computing%20Krippendorff's%27s%20Alpha-Reliability.pdf?utm_source=chatgpt.com
- 18 **Ethics |**
https://aoir.org/ethics/?utm_source=chatgpt.com
- 19 **RightsStatements.org**
https://rightsstatements.org/en/?utm_source=chatgpt.com
- 20 **Fair dealing exceptions**
https://www.alrc.gov.au/publication/copyright-and-the-digital-economy-ip-42/fair-dealing-exceptions/?utm_source=chatgpt.com
- 21 **social-media-research-ethics-policy-note-university- ...**
https://www.utwente.nl/en/bms/research/forms-and-downloads/social-media-research-ethics-policy-note-university-sheffield.pdf?utm_source=chatgpt.com
- 22 **AIATSIS Code of Ethics for Aboriginal and Torres Strait ...**
https://aiatsis.gov.au/sites/default/files/2020-10/aiatsis-code-ethics.pdf?utm_source=chatgpt.com

24 **Documentation: 18: 5.9. Row Security Policies**

https://www.postgresql.org/docs/current/ddl-rowsecurity.html?utm_source=chatgpt.com

25 **Warming stripes**

https://www.climate-lab-book.ac.uk/2018/warming-stripes/?utm_source=chatgpt.com

26 **WMO Statement on the Status of the Global Climate in 2003**

https://digitallibrary.un.org/record/3893720/files/966_en.pdf?utm_source=chatgpt.com

27 **Extreme Heat**

https://wmo.int/themes/extreme-heat?utm_source=chatgpt.com

28 **Documentation: 18: Chapter 25. Backup and Restore**

https://www.postgresql.org/docs/current/backup.html?utm_source=chatgpt.com